

Written explanation of the idempotency mechanism and why it matters in payroll.

Idempotency is a mechanism that ensures the same payroll transaction is processed only once, even if AWS Lambda retries execution or Amazon SQS delivers duplicate messages.

In our payroll system, each payroll record is uniquely identified using a combination of `run_id` and `employee_id`. Before creating a payroll record, the Worker Lambda attempts to insert the payroll data into the `payroll_runs` DynamoDB table using a conditional write operation.

If a record with the same `run_id` and `employee_id` already exists, DynamoDB rejects the operation and the Lambda function skips processing for that employee. As a result, duplicate salary calculations, duplicate PDF generation, and duplicate email notifications are prevented.

For example, if the payroll run ID is "231f0259-3265-4dc3" and the employee ID is "EMP003", the system creates a unique payroll record for that employee in that specific payroll cycle. If the same SQS message is processed again due to a retry, the conditional check fails and the employee is not processed a second time.

This mechanism is critical in payroll systems because duplicate processing can result in incorrect payroll records, multiple payslip deliveries, financial inconsistencies, and audit issues. By using DynamoDB conditional writes, the system guarantees that each employee is processed only once per payroll run, ensuring reliability, accuracy, and fault tolerance.